

Restaurant Survival Classification

Algorithms, Selection Process, Expected Results

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The Problem

- Binary classification: will a restaurant **close** (`status_closed = 1`)?
- Features: Google-Places metadata, review dynamics, local competition, demographics
- **Class imbalance**: only ~10% positives in the training set
- **Scoring metric: Balanced Accuracy (BA)** — accuracy of each class, averaged
- Every modelling decision below is driven by BA, not plain accuracy

Algorithms Considered (ML1 scope)

Screened with **5-fold stratified CV** on the engineered feature set:

Algorithm	Why considered
L2 Logistic Regression	Linear baseline, calibrated, weighted
L1 Logistic Regression	Embedded feature selection
KNN ($k = 25$, dist-weighted)	Non-parametric sanity check
Decision Tree (depth 8)	Interpretable non-linear baseline
Random Forest (500 trees)	Bagged trees — non-linear interactions

Dropped early: KNN (suffers in high-dim numeric space) and the single tree (underfits).

Advanced: L1 / L2 Logistic and Random Forest.

Feature Engineering (~40 new features)

- **Log transforms** on heavy-tailed counts (reviews, POIs, restaurants)
- **Recency shares**: 1m / 3m / 6m / 12m of total reviews
- **Momentum / burst**: e.g. momentum_3m_to_12m
- **Local competition**: reviews_per_local_restaurant, restaurant_density, poi_density_500m
- **Rating interactions**: rating_x_popularity, rating_x_age, rating_consistency
- **Missingness indicators** for high-NA columns

Competition (catch_) and recency features were among the most influential.*

Selection Process — Hyperparameter Tuning

Tuned with **out-of-fold (OOF) CV** — never on the test set.

- **L2 Logistic:** grid over $C \in \{0.1, 0.2, 0.4, 1.0\}$ and class-weight multiplier $\in \{0.5 \dots 1.0\}$
 - Best: $C = 0.40$, multiplier = 0.60 — **OOF BA** ≈ 0.677
- **Random Forest:** grid over `n_estimators`, `max_features`, `min_samples_leaf`
- **Decision threshold** tuned on validation probabilities (not fixed at 0.5)

Selection Process — Ensembling

The linear and tree models make **different errors** $\Rightarrow$ ensemble.

Ensemble	Mechanism	Val BA*
Soft Voting	Average predict_proba	mid
Stacking	Logistic meta-learner	mid
Probability blend	Tuned convex combination	best

Winner: $0.775 \times \text{elastic-net} + 0.225 \times \text{bagged trees}$

- Elastic-net (`l1_ratio = 0.10`): embedded selection + L2 stability
- BaggingClassifier: 400 trees, `min_samples_leaf = 20`
- **Threshold** ≈ 0.43 (tuned on hold-out validation set)

Why the Blend?

- **Elastic-net** handles the $\sim 10\%$ imbalance via explicit `class_weight` and selects features through its L1 component.
- **Bagged decision trees** add non-linear signal on competition and recency features without overfitting (400 bootstrap replicates).
- **Probability blend** beat soft voting and stacking on the hold-out validation set.
- **Threshold tuning** recovers BA points lost to class imbalance, without leakage.
- Entirely within the ML1 syllabus — no boosting libraries.

Expected Results

All numbers measured **without touching the Kaggle test set**:

Setup	BA
5-fold CV, single models, threshold 0.5	0.62 – 0.66
Tuned L2 logistic, OOF, optimal threshold	≈ 0.677
Enet + Bag blend, hold-out validation	0.68 – 0.69
Honest 20% internal test (one look)	0.67 – 0.69

Expected Kaggle test BA: high 0.6s. Blend lift over best single model: **+0.005 to +0.01 BA.**

Summary

- Considered: L1/L2 Logistic, KNN, Decision Tree, Random Forest.
- Selected via 5-fold CV $\rightarrow$ OOF hyperparameter grids $\rightarrow$ ensemble comparison.
- **Final model:** Elastic-net + Bagged Trees probability blend (0.775/0.225), threshold ≈ 0.43 .
- **Expected Balanced Accuracy: 0.68 – 0.69.**
- Honest selection (OOF only) $\Rightarrow$ reported BA is a realistic estimate of test performance.

Thank you — questions?